Sheet (4)

Multiple Choice Questions:

1) Which of the following quantities is transferred during wave propagation?

a. Energy **b.** Speed **c.** Matter

2) If you double the frequency of a vibrating object, what happens to the period?

a. doubled **b.** Stay constant **c.** Fall to the half

3) Mechanical waves could be transferred through

a. Space **b.** solid, liquid, gas **c.** both i and ii

4) Which of the following is an example of S.H.M.

a. Pendulum motion **b.** mass connected to a stretched spring **c.** both i and ii

5) The wave speed is …. …to the angular frequency

a. inversely proportional **b.** directly proportional **c.** Nonrelative

6) Which of the following isn't an example of electromagnetic waves?

a. radio-waves **b.** ultrasound waves **c.** microwaves

Problems

7) Extreme waves along ocean waters, sometimes referred to as freak waves or rogue waves, are a focus of much research and study among scientists. Several merchant ships report rogue waves which are estimated to be 25 meters high and 26 meters long (successive crests). If these waves travel at speeds of 6.5 m/s, determine

(a) The frequency and the period

(b) How many waves pass through a distance equivalent to 2?

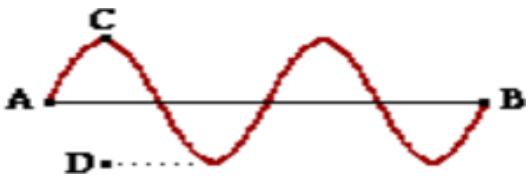
8) A sinusoidal wave is traveling through a rope. The oscillator generates the waves and generates 40 vibrations per 30 seconds. A given crest travels through 425 cm along the rope n 10 seconds.

Calculate the wavelength of the wave.

9) The distance between two successive crests is 1.2 m. Eight crests pass through a given point every 12 seconds.

Calculate the wave speed.

10) A wave with a frequency of 12.3 Hz is traveling from left to right across a rope as shown in the diagram at the right Positions A and B in the diagram are separated by a horizontal distance of 42.8 cm. Positions C and D in the diagram are separated by a vertical distance of 12.4 cm. Determine the



(a) amplitude, (b) wavelength, (c) period (d) Speed of this wave

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11) Write an expression that describes the pressure variation as a function of position and time for a sinusoidal sound wave in air at $T = 20^{o}C$ given that the distance between two successive compression is 0.1 m and the maximum amplitude for pressure = 0.2 Pa.

12) Calculate the pressure amplitude of a 2000 Hz sound wave in air at 20 ^{o}C if the displacement amplitude is equal to 2×10^{-8} m if the density of the air is 1.3 kg/m^3.

13) A Sound level 2 meters from a source is measured to be 90 dB. How far away must a person be to measure a sound level of 50 dB?

14) The area of a typical eardrum is $5 * 10^{-5}\ m^2$. Calculate the average sound power incident on an eardrum at the threshold of pain, which corresponds to an intensity of 1.0 W/m^2. How much energy is transferred to the eardrum to the eardrum exposed to this sound for 1 minute?

Multiple Choice Questions:

15) ……...is when two waves that are in the same place and cancel each other out.

a. Constructive interference **b.** Destructive Interference **c.** Polarization

16) The wave equation for a standing wave on a string fixed at both ends is $y\ (x,\ t) = 24$ $\cos(3.0x)\ \cos(2t)$ The speed of the traveling waves that result in this Standing

a. 0.67 m/s. **b.** 1.5 m/s **c.** 8.0 m/s.

17) A person standing in a swimming pool appears to have short legs because of…...

a. Light Diffraction **b.** Light Refraction **c.** Light Reflection

Problems

18) Two waves are travelling in the same direction along a stretched string. The waves are 90 out of phase. Each wave has an amplitude of 4cm. Find the amplitude of the resultant wave.

19) Find an expression for the displacement "y" produced when there are two waves present with displacements: $y1 = 4\ sin\ (5\ x - 2\ t)$ $y2 = 4\ sin\ (5\ x - 2\ t)$

20) Two harmonic waves are described by $y1 = (5m)\ sin\ (\pi\ (4x - 1200\ t))$ $y2 = (5\ m)$ $sin\ (\pi\ (4x - 1200\ t - 0.25)$ Where x in meters and t in seconds (a) What is the amplitude of the resultant wave? (b) What is the frequency of the resultant wave? (c) What is the wavelength of the resultant wave?